

8(a)	<table><tr><td>location of Na⁺ / K⁺ pumps</td><td>P</td><td rowspan="4">;</td><td rowspan="4">Ignore Q</td></tr><tr><td>location of cotransport proteins</td><td>S</td></tr><tr><td>structures through which glucose molecules pass</td><td>P + S</td></tr><tr><td>prevents leakage of fluid</td><td>R</td></tr></table>	location of Na ⁺ / K ⁺ pumps	P	;	Ignore Q	location of cotransport proteins	S	structures through which glucose molecules pass	P + S	prevents leakage of fluid	R	4
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8(b)	any seven from: 1 increase in ADH released ; Ignore produced A secreted 2 (ADH released from) posterior pituitary ; 3 ADH binds to receptors ; 4 in, cell surface / basal, membrane, of collecting duct (cells) ; 5 cAMP / secondary messenger, produced ; 6 <i>ref. to</i> enzyme cascade / signal amplification / described ; 7 vesicles (with aquaporins) move to / fuse with, cell surface / luminal, membrane ; 8 more aquaporins in the, cell surface / luminal, membrane ; 9 increase in (cell / membrane / collecting duct) permeability to water ; 10 (more) water moves out of, lumen / collecting duct / filtrate, into, cell / tissue fluid / blood ; 11 by, osmosis / down the water potential gradient ; 12 (more) concentrated / small(er) <u>volume</u>, urine produced ;	7										